



Semester 2 Examinations 2012/ 2013

Exam Code(s) 3BA1, 3BS4, 3BMS2, 4BME1, 4BS3, 4FM2, 4BS9, 1HDS1
Exam(s) 3rd Arts, 3rd Science, 4th Science, 4th Financial Mathematics and Economics, 4th Mathematics and Education, H. Dip. (Appl. Sc)

Module Code(s) MP491
Module(s) Non-linear Systems

Paper No. 1
Repeat Paper

External Examiner(s) Professor G. Saccomandi
Internal Examiner(s) Dr P. T. Piironen
Professor M. Destrade

Instructions: For full marks answer question 1 in Section A **and** TWO questions in Section B correctly.
All questions carry the same weight.

Duration 2hrs
No. of Pages 4
Department(s) Applied Mathematics
Course Co-ordinator(s) Dr P. T. Piironen

Requirements:

MCQ
Handout
Statistical/ Log Tables Yes
Cambridge Tables
Graph Paper
Log Graph Paper
Other Materials

Section A - Answer all questions (a)-(d). (5 Marks/question)

1. (a) Consider the nonlinear ordinary differential equation

$$\frac{dx}{dt} = 2x - x^2 + 3.$$

Find all equilibrium points and determine if they are stable or not.

- (b) Consider the linear system of ordinary differential equations

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ -2 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

Classify the equilibrium point $(x, y) = (0, 0)$ and sketch the phase-plane portrait.

- (c) Consider the system of nonlinear ordinary differential equations

$$\begin{aligned} \frac{dx}{dt} &= (x+2)(y-2), \\ \frac{dy}{dt} &= (x-1)(y+2). \end{aligned}$$

Find all equilibrium points and classify them.

- (d) Consider the non-linear map

$$x_{n+1} = x_n - (x_n)^2 + 4.$$

Find all fixed points and determine if they are stable or not.

Section B - Answer any two questions.

2. (a) Consider the one-parameter family of nonlinear ordinary differential equations

$$\frac{dx}{dt} = x^2 + (\mu^2 - 5)x + 4 - \mu^2,$$

where μ is a real parameter. Show that one of the equilibrium points is given by $x = 1$ and find the remaining equilibrium points. Sketch the bifurcation diagram, and indicate the behaviour of the non-equilibrium solutions. Find all bifurcation points and classify them.

10 Marks

- (b) Consider the system of ordinary differential equations

$$\begin{aligned}\dot{x} &= 2x + y + x^2 - xy, \\ \dot{y} &= 3\mu x + \mu y - x^2 y.\end{aligned}$$

Find the value of μ for which a Hopf bifurcation occurs at the origin $(x, y) = (0, 0)$. How does the nature of the equilibrium point change at the Hopf bifurcation?

10 Marks

3. Consider the system of ordinary differential equations

$$\begin{aligned}\frac{dx}{dt} &= (x - 2)(y + 2), \\ \frac{dy}{dt} &= (x + 1)(y - 1).\end{aligned}$$

- (a) Locate and classify all equilibrium points in the associated phase plane.

4 Marks

- (b) Identify, and sketch separately, the isoclines corresponding to $k = -1, 0, 1, \infty$, where

$$\frac{dy}{dx} = k.$$

12 Marks

- (c) By using all four isoclines in part (b), sketch a sufficient number of phase plane curves to illustrate the phase-plane portrait for this system.

4 Marks

4. Consider the second-order nonlinear ordinary differential equation

$$\ddot{z} + z^2 + 3z = -2.$$

- (a) Show that the second order nonlinear ordinary differential equation can be written as

$$\begin{aligned}\dot{x} &= y, \\ \dot{y} &= -x^2 - 3x - 2.\end{aligned}$$

2 Marks

- (b) Show that the ODE system in part (a) is Hamiltonian and find the Hamiltonian function.

4 Marks

- (c) Find all equilibrium points and classify them.

4 Marks

- (d) Use the Hamiltonian function found in part (b) to sketch the phase-plane portrait.

10 Marks